Huawei Access Node BAS-ACC-05 Routing Protocol Flap SOP (ALARM_9778)

Node: BAS-ACC-05 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_9778

Description

This Standard Operating Procedure (SOP) outlines the steps to diagnose and resolve a Routing Protocol Flap failure on the Huawei Access Node BAS-ACC-05, located at the Basavanagudi POP. A Routing Protocol Flap, denoted by Alarm Code ALARM_9778, occurs due to the churning of BGP routes, loss of OSPF adjacency, or excessive ISIS LSP flooding. This failure domain is particularly pertinent to the Access role, which is responsible for establishing and maintaining routing sessions with neighboring nodes. The Huawei NE Series routers utilize a complex interplay of protocols, including BGP, OSPF, and ISIS, to facilitate routing decisions. The flapping of these protocols can be attributed to various factors, including misconfigurations, hardware issues, or network congestion. The root cause of the failure must be identified and addressed to prevent further disruptions to the network. This SOP outlines the necessary diagnostic commands, resolution steps, verification commands, and rollback procedures to ensure a swift and efficient resolution.

Pre-requisite Checks

- check: Verify that the Basavanagudi POP is accessible and that all necessary cables are securely connected., status: Complete

- check: Check the status of the redundant link at the Basavanagudi POP and ensure it is operational., status: Complete

- check: Dispatch remote hands to the Basavanagudi POP to facilitate access to the Huawei NE Series router., status: Complete

Potential Root Causes

Cause	Probability
Misconfiguration of BGP routing policies	High
Hardware issue with the NE Series router's CPU or memory	Medium
Network congestion or oversubscription	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp routing-policy, Expected Output: Displayed BGP routing policies, Description: This command retrieves the BGP routing policies configured on the router. The output will indicate if there are any misconfigured policies that may be contributing to the flapping.
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```
Command: display ospf neighbor, Expected Output: Displayed OSPF neighbor information, Description: This command retrieves the OSPF neighbor information, including the states of the adjacencies. The output will indicate if there are any issues with the OSPF adjacency.

Command: display isis lsp, Expected Output: Displayed ISIS LSP information, Description: This command retrieves the ISIS LSP information, including the LSP flooding status. The output will indicate if there are any issues with the ISIS LSP flooding.
```

Resolution Steps

Step 1: Step: Verify and correct BGP routing policies, Command: undo routing-policy bgp , Description: If misconfigured BGP routing policies are identified, this command will undo the policies to prevent further flapping.

Step 2: Step: Restart the OSPF process, Command: restart ospf, Description: If issues with the OSPF adjacency are identified, restarting the OSPF process may resolve the issue.

Step 3: Step: Clear ISIS LSP flooding, Command: clear isis lsp, Description: If issues with the ISIS LSP flooding are identified, clearing the LSP flooding may resolve the issue.

Post-Check Verification

```
Command: display bgp routing-policy, Expected Output: Displayed BGP routing policies without misconfigurations, Description: This command verifies that the BGP routing policies have been corrected.

Command: display ospf neighbor, Expected Output: Displayed OSPF neighbor information with stable adjacencies, Description: This command verifies that the OSPF adjacency has stabilized.

Command: display isis lsp, Expected Output: Displayed ISIS LSP information without LSP flooding, Description: This command verifies that the ISIS LSP flooding has been cleared.
```

Roll-back Procedures

- Step: Restore the original BGP routing policies, Command: apply routing-policy bgp , Description: If the fix fails, this command will restore the original BGP routing policies.

- Step: Restart the OSPF process, Command: restart ospf, Description: If the fix fails, restarting the OSPF process may resolve the issue.

- Step: Re-run ISIS LSP flooding, Command: run isis lsp, Description: If the fix fails, re-running the ISIS LSP flooding may resolve the issue.